Dynamic Threshold Detection for Methylation-Based Binary Encoding

Decoding ASCII-to-Binary Conversion in DNA Methylation Patterns

BioCompute $\times$ IIT-Delhi

Meeting Progress Report

May 16, 2026

Outline

Research Objective

Primary Goal

Decode encrypted ASCII-to-binary messages stored in DNA methylation patterns

Encoding Scheme:

- Message: ``EpiC''
- ASCII $\rightarrow$ 8-bit binary
- Stored in CpG methylation (36 positions)
- $1 =$ methylated, $0 =$ unmethylated

Challenge:

- Variable methylation levels
- Signal-to-noise separation
- Optimal threshold detection
- Q-score filtering impact

Key Question: Can dynamic thresholding based on signal separation enable reliable binary decoding?

Sample Overview

Sample	Expected	Q-scores	Status
EpiC_26_3_26	EpiC	Q0, Q5, Q10, Q15, Q20	Strong Signal
EpiC_25_2_26	EpiC	Q0, Q5, Q10, Q15, Q20	Perfect Decoder
EpiC_26_2_26	EpiC	Q0, Q5, Q10, Q15, Q20	Weak Signal
EpiC_31_3_26	EpiC	Q0, Q5, Q10, Q15, Q20	Good Signal

- **Total analyses:** 20 (4 samples × 5 Q-scores)
- **Data source:** Oxford Nanopore 10.4.1 MINion sequencing with Modkit
- **Reference:** 36 CpG positions mapped to custom oligo sequences

Q-Score Impact Analysis

Key Finding: Q-score filtering has significant impact on signal quality

Sample	Q0	Q5	Q10	Q15	Q20
EpiC_26_3_26	96.9%	96.9%	96.9%	81.2%	59.4%
EpiC_25_2_26	100%	100%	100%	100%	100%
EpiC_26_2_26	59.4%	59.4%	59.4%	59.4%	59.4%
EpiC_31_3_26	96.9%	96.9%	96.9%	78.1%	59.4%
Average	**88.3%**	**88.3%**	**88.3%**	**79.7%**	**69.5%**

Table: Decoding accuracy across Q-scores

Recommendation

Use Q0, Q5, or Q10 for maximum coverage and accuracy. Q15+ cause coverage loss.

Dynamic Threshold Detection

Core Concept

Automatically determine optimal threshold based on **signal separation** between expected 0s and 1s

Algorithm:

1. Parse BED file for methylation data
2. Separate positions by expected bit (0 or 1)
3. Calculate distribution statistics
4. Measure signal separation:
   - Fold-change (mean & median)
   - Cohen's *d* (effect size)
5. Determine optimal threshold
6. Classify confidence level

Key Metrics:

- **Fold-change:** $FC = \frac{\mu_1}{\mu_0}$
- **Cohen's d:** $d = \frac{\mu_1 - \mu_0}{\sigma_{pooled}}$
- **Significance:** $\mathbf{FC} \geq 3.0 + d \geq 2.0$

Fold-Change Analysis: Overall Results

Sample	Min FC	Max FC	Mean FC	Q for Max	Viable
EpiC_25_2_26	18.01x	23.51x	20.37x	Q20	✓
EpiC_26_2_26	7.20x	8.95x	7.60x	Q20	!
EpiC_26_3_26	5.58x	5.59x	5.58x	Q0	✓
EpiC_31_3_26	17.67x	32.62x	21.76x	Q15	✓

Key Finding

Signal separation ranges from 5.6x to 32.6x fold-change, but not all samples are practically viable for decoding

Statistical vs. Practical Significance

Critical Distinction: High fold-change does not always mean practical viability

Practical Viability Criteria

- Mean methylation for "1s" $\geq$ 10%
- Absolute difference between "1s" and "0s" $\geq$ 5%
- Signal must be distinguishable from technical noise

Case Study: EpiC_26_2_26

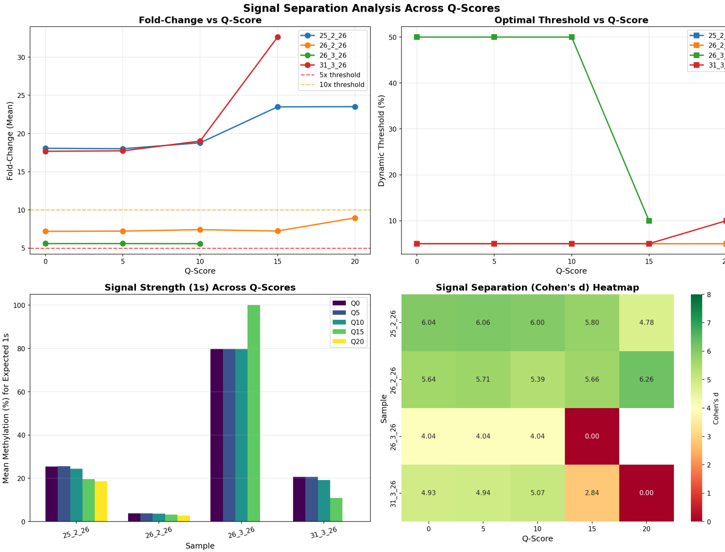
- Fold-change: 7.2x - 8.95x (✓)
- Cohen's *d*: 5.39 - 6.26 (✓)
- Mean "1s": 2.8 - 4.3% (!)
- Mean "0s": 0.3 - 0.5% (!)

Verdict:

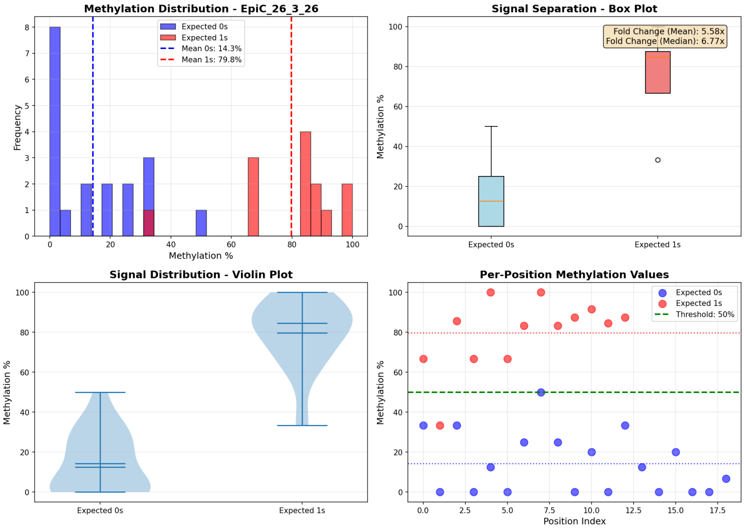
Statistically significant but **NOT practically viable**

Absolute methylation levels too low to distinguish from background noise in real-world applications.

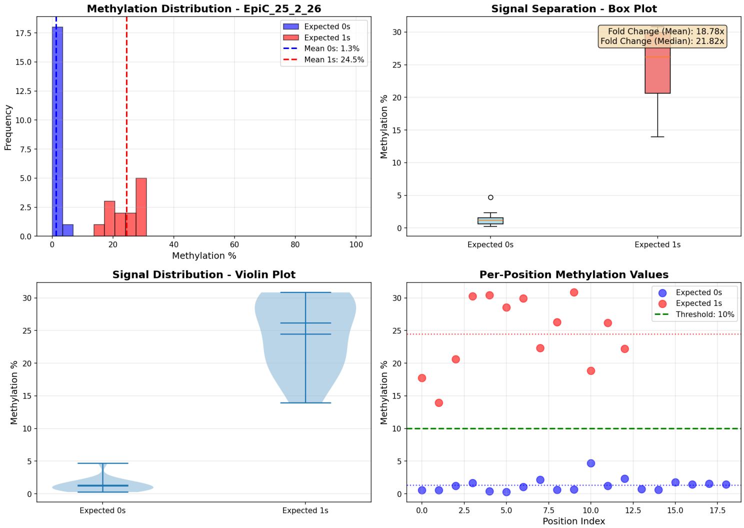
Fold-Change Across Q-Scores



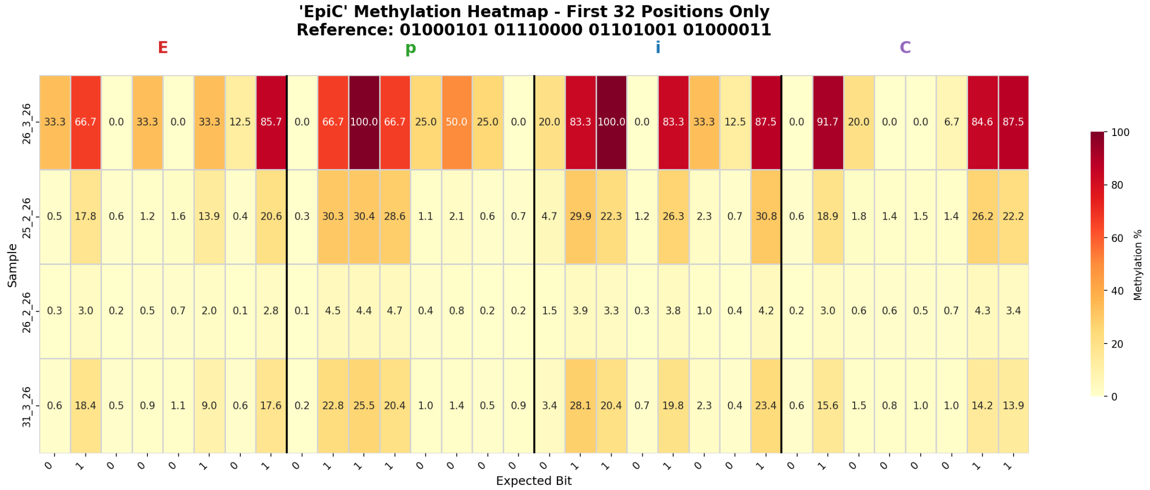
Signal Distribution Example: EpiC_26_3_26



Signal Distribution Example: EpiC_25_2_26



Methylation Heatmap: 'EpiC' Encoding



Key Finding 1: Signal Separation Quality

Primary Discovery

Dynamic thresholding reveals **5.6x to 32.6x fold-change**, but only **3 out of 4 samples** are practically viable

Implications:

- Far exceeds initial claim of "4-7 fold" difference
- **BUT:** Statistical significance $\neq$ practical viability
- Absolute methylation levels matter as much as fold-change
- Signal quality varies significantly by sample

Practically Viable (✓):

- EpiC_25_2_26: 18-24x, 19-26%
- EpiC_31_3_26: 18-33x, 19-22%
- EpiC_26_3_26: 5.6x, 80%

Not Viable (!):

- EpiC_26_2_26: 7-9x, 2.8-4.3%
- Good fold-change
- **But** absolute signal too weak

Key Finding 2: Q-Score Effects

Q-score impact varies by sample:

Stable Samples:

- EpiC_26_3_26: 5.58-5.59x (all Q)
- EpiC_26_2_26: 7.2-9.0x
- Minimal Q-score dependency
- Consistent performance

Variable Samples:

- EpiC_25_2_26: 18-24x
- EpiC_31_3_26: 18-33x
- Q15/Q20 increase fold-change
- But reduce coverage

Recommendation:

Q0-Q10 optimal for most samples
Q15+ causes signal degradation

Mechanism:

- High Q filters low-quality reads
- Reduces 0-signal more than 1-signal
- Increases apparent fold-change
- But may lose real signal

Key Finding 3: Dynamic vs. Fixed Thresholds

Sample	Fixed	Dynamic	Improvement
EpiC_26_3_26	50% (manual)	50.0%	Confirmed
EpiC_25_2_26	10% (manual)	5.0%	Optimized
EpiC_31_3_26	10% (manual)	5.0%	Optimized
EpiC_26_2_26	10% (manual)	5.0%	Optimized

Statistical Significance vs. Practical Viability

All samples show statistically significant signal separation, but not all are practically viable:

Sample	Cohen's d	Fold-Change	Mean "1s"	Statistical	Practical
EpiC_25_2_26	4.78 - 6.06	18.0 - 23.5x	19-26%	✓	✓
EpiC_26_2_26	5.39 - 6.26	7.2 - 9.0x	2.8-4.3%	✓	!
EpiC_26_3_26	4.04	5.6x	80%	✓	✓
EpiC_31_3_26	2.84 - 5.07	17.7 - 32.6x	19-22%	✓	✓

Cohen's d Interpretation

- $d = 0.8$: Large effect
- $d > 2.0$: Very large effect
- All samples $d > 2.8$

Practical Viability

- Mean "1s" $\geq 10\%$
- Absolute diff $\geq 5\%$
- **EpiC_26_2_26 fails this**

Conclusions

1. **Statistical significance does not guarantee practical viability**
   - 5.6x - 32.6x fold-change, Cohen's $d = 2.84$ - 6.26
   - **BUT:** Only 3 out of 4 samples are practically viable
   - EpiC_26_2_26: Good fold-change but absolute signal too weak (¡10%)
   - **Key insight:** Must consider absolute methylation levels, not just ratios

2. **Q-score filtering should be conservative (Q0-Q10)**
   - Q15+ reduces coverage and degrades weak signals
   - Q10 provides best balance
   - High Q-scores increase fold-change but reduce practical signal

3. **Dynamic threshold detection enables practical viability assessment**
   - Adapts to sample-specific characteristics
   - Provides both statistical and practical confidence metrics
   - Flags samples with insufficient absolute signal strength

Technical Implementation

Codes:

1. `dynamic_threshold_detector.py`
   - Automatic threshold detection
   - Batch processing capability
   - **Practical viability assessment**
   - Distinguishes statistical from practical significance
   - JSON output for reproducibility
2. `signal_strength_analysis.py`
   - Distribution analysis
   - Fold-change calculations
   - Visualization generation
3. `comprehensive_qscore_analysis.py`
   - Multi-sample, multi-Q-score analysis
   - Summary statistics
   - Comparative visualizations

Thank You

Questions?

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Additional Data: Complete Results Table I

Sample	Q	Thresh	Mean 0s	Mean 1s	FC	Cohen d	Conf	Viable
EpiC_25_2_26	0	5.0%	1.41%	25.49%	18.06x	6.04	HIGH	✓
EpiC_25_2_26	5	5.0%	1.42%	25.60%	18.01x	6.06	HIGH	✓
EpiC_25_2_26	10	5.0%	1.30%	24.47%	18.78x	6.00	HIGH	✓
EpiC_25_2_26	15	5.0%	0.84%	19.69%	23.48x	5.80	HIGH	✓
EpiC_25_2_26	20	5.0%	0.79%	18.54%	23.51x	4.78	HIGH	✓
EpiC_26_2_26	0	5.0%	0.52%	3.75%	7.20x	5.64	HIGH	!
EpiC_26_2_26	5	5.0%	0.52%	3.77%	7.22x	5.71	HIGH	!
EpiC_26_2_26	10	5.0%	0.49%	3.64%	7.41x	5.39	HIGH	!
EpiC_26_2_26	15	5.0%	0.44%	3.20%	7.23x	5.66	HIGH	!
EpiC_26_2_26	20	5.0%	0.32%	2.84%	8.95x	6.26	HIGH	!
EpiC_26_3_26	0	50.0%	14.28%	79.77%	5.59x	4.04	HIGH	✓
EpiC_26_3_26	5	50.0%	14.28%	79.77%	5.59x	4.04	HIGH	✓
EpiC_26_3_26	10	50.0%	14.30%	79.77%	5.58x	4.04	HIGH	✓
EpiC_31_3_26	0	5.0%	1.17%	20.64%	17.67x	4.93	HIGH	✓
EpiC_31_3_26	5	5.0%	1.16%	20.63%	17.73x	4.94	HIGH	✓
EpiC_31_3_26	10	5.0%	1.01%	19.16%	19.01x	5.07	HIGH	✓
EpiC_31_3_26	15	5.0%	0.33%	10.88%	32.62x	2.84	HIGH	✓